

Structural Observables Architecture Manual

RDF, Coordination, Neighbor Angles, Atomic Connectivity, and General Structural Validation

mdstats project

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Purpose and status

This manual owns the general static structural observables in `mdstats.analysis`. It covers pair radial distributions, coordination states, neighbor-angle distributions, and atomic connectivity. These observables are usable for crystalline solids, amorphous solids, liquids, surfaces, and interfaces when their selections and neighbor definitions are scientifically appropriate.

The following public implementations are complete and regression-tested:

- `compute_pair_rdf` and `RDFResult` feature helpers;
- `compute_coordination_distribution`;
- `compute_bond_angle_distribution`;
- `compute_atomic_connectivity` with distance, hysteretic, reference, and explicit definitions;
- catalog-derived atomic connectivity statistics in `mdstats.analysis.topology_statistics`;
- `compute_local_structure_features` with chemistry-scaled smooth connectivity, radial projections, angular moments, local-density proxy, and local q_4/q_6 orientational order for selection-grade use.

The DATA9A7b focused gate adds analytic, invariance, periodicity, missing-value, and complexity-budget tests for the local-structure kernel. The standardized observable-call facade remains separate; the local selection kernel is not automatically registered as a physical-validation observable.

Phase-aware MLFF consumption

MLFF-DATA9A7c may choose which selection-grade local-structure feature families are exposed for a declared phase or geometry. That policy filtering does not change ownership: `mdstats.analysis.local_structure` calculates the complete numerical local environment under an explicit policy, while the MLFF branch selects and aggregates authorized columns for dataset coverage. Phase profiles must not redefine RDF, coordination-distribution, angle-distribution, topology, or physical validation semantics.

Ownership boundary

The analysis modules own:

1. mathematical definitions and normalization;
2. periodic geometry and neighbor semantics;
3. result dataclasses and scientific metadata;
4. warnings, failure modes, and numerical validation;
5. module-specific plotting and export where already implemented.

The MLFF branch may select, parameterize, invoke, pair, and record these analyses. It must not duplicate their algorithms, reinterpret their arrays, or claim that a call alone constitutes scientific acceptance.

The stable bridge is analysis-owned:

```
from mdstats.analysis import (
    ObservableAnalysisCall,
    ObservableAnalysisRecipe,
    execute_observable_recipe,
)
```

`mdstats.training_data` owns only paired reference/MLFF orchestration through `MLFFObservableValidationPlan` and `run_mlff_observable_validation`.

Shared neighbor and geometry contract

RDF, coordination, angle, and connectivity results depend on a declared neighborhood policy. The policy is part of the scientific observable and must not be treated as a plotting option.

The underlying periodic neighbor subsystem owns:

- exact triclinic minimum-image geometry;
- strict cutoff semantics;
- dense, cell-list, and Verlet-cache execution;
- deterministic CSR neighbor results;
- deformation-aware candidate reuse where explicitly enabled;
- backend diagnostics without changing accepted physical pairs.

The controlling architecture is `periodic_neighbor_search_architecture.{md,pdf}`.

Selection-grade local-structure kernel

`mdstats.analysis.local_structure` owns a reusable per-atom structural feature kernel. It was introduced for MLFF data selection, but its geometry and normalization remain analysis responsibilities so that other branches may reuse it without importing `mdstats.training_data`.

For pair separation r_{ij} , a continuous weight is generated from the normalized distance $r_{ij}/(R_i + R_j)$ using a cosine switching interval. The resulting weighted degree is a smooth coordination proxy. The kernel also returns nearest-neighbor and weighted-distance summaries, Gaussian radial projections, a Gaussian local-density proxy, weighted neighbor-species entropy, Legendre angular moments, and weighted spherical-harmonic q_4 and q_6 .

The result is invariant to rigid translation, global rotation, and atom ordering. Undefined angular moments use an explicit missing mask rather than a fabricated number. The complete radius, switching, radial-basis, angular-order, and work budget policy is serialized with the result lineage.

This first implementation is intentionally transparent and dense. It fails closed when $N_c N$ exceeds the declared pair-work budget. A later optimization may use the periodic cell-list or Verlet backend while retaining the same public contract.

These features are not, by themselves, validation-grade RDFs, angle distributions, integer coordination states, or phase classifiers. The MLFF branch may aggregate them for selection, but interpretation remains outside the kernel.

Pair radial distribution

`compute_pair_rdf` owns partial pair histograms, shell-volume normalization, and cumulative coordination. `RDFResult` additionally owns auditable smoothing, first-peak, first-minimum, and first-shell coordination helpers.

RDF is suitable for broad structural comparison, but it is not a complete local environment representation. Similar RDFs can hide different angular, connectivity, or medium-range structures.

Coordination distributions

`compute_coordination_distribution` returns the authoritative integer per-center, per-frame coordination matrix together with distributions and summary statistics. The cutoff may be explicit, supplied by a pair-cutoff registry, or derived from a compatible RDF minimum.

A coordination cutoff is a model definition. For liquids and thermally active systems, users should consider sensitivity to the cutoff and use hysteresis or smooth coordination in future extensions when state switching is the target.

Neighbor-angle distributions

`compute_bond_angle_distribution` computes species-resolved A-B-C neighbor angles, supports coordination filters, and exposes angle-, center-, and frame-weighted statistics. The term *neighbor angle* is preferred unless a chemical bond definition has been established independently.

Atomic connectivity

`compute_atomic_connectivity` owns discrete weighted-by-definition graph states for fixed-population frame collections. Supported policies include:

- instantaneous strict-distance connectivity;
- two-threshold hysteretic trajectory connectivity;
- reference-state discovery/formation/retention connectivity;
- externally supplied explicit connectivity.

Connectivity is not universal chemical bonding. Its definition, scope, pair cutoffs, frame ordering, and persistence policy are mandatory provenance.

The topology-statistics branch, not the MLFF branch, owns graph-state occupancy, degree and pair distributions, contact persistence, transitions, and residence statistics.

Standardized observable calls

`mdstats.analysis.observable_validation` provides a registry and immutable JSON-safe recipes. It delegates to the owner function and preserves native result objects.

Implemented structural call IDs are:

Observable ID	Owner
structure.rdf	mdstats.analysis.rdf.compute_pair_rdf
structure.coordination	mdstats.analysis.coordination.compute_coordination_distribution
structure.bond_angle	mdstats.analysis.bond_angle.compute_bond_angle_distribution
topology.atomic_connectivity	mdstats.analysis.atomic_connectivity.compute_atomic_connectivity
topology.atomic_statistics	mdstats.analysis.topology_statistics.compute_atomic_connectivity_statistics

The facade may decode simple serialized pair-cutoff registries and distance, hysteretic, or reference connectivity definitions. More specialized framework and ring calls remain direct APIs owned by their manuals.

Physical validation role

For MLFF validation, these analyses are usually run on matched reference and model-generated configurations or trajectories. The call parameters must match:

- atom/species/group selection;
- thermodynamic condition;
- cell and periodicity semantics;
- frame window and stride;
- neighbor and cutoff definition;
- averaging and normalization.

Comparison metrics and acceptance tolerances are separate policy. For example, an RDF comparison may use pointwise differences, integrated absolute error, peak displacement, or shell-coordination differences. This manual does not choose one universal tolerance.

Missing general structural observables

The following additions are recommended, but are not implemented in this package revision:

Static structure factor

Add total and partial $S(q)$ from positions or from compatible RDFs, including finite-cell and scattering-weight provenance. This is the highest-priority missing general validation observable for liquids and amorphous solids.

Bond-orientational order

A selection-grade local q_4/q_6 implementation now exists in `local_structure`. A full validation observable is still missing: it should add local and global distributions, neighbor-averaged variants, explicit phase classification policies, uncertainty summaries, and standardized observable call/result schemas. These are useful for crystallinity, freezing, polymorph discrimination, and amorphous local order.

Local volume and packing

The local-structure kernel includes a Gaussian number-density proxy, but it does not provide geometric local volume or free volume. Add Voronoi or another declared local-volume estimator, local composition, packing anisotropy, and free-volume summaries. Robust periodic triclinic support and degeneracy handling are required before release.

Crystalline local classifiers

Polyhedral-template matching, adaptive common-neighbor analysis, local lattice orientation, and local elastic strain should form an optional crystalline subbranch. They must not become universal defaults.

Dihedral and molecular internal coordinates

Add generic dihedral distributions and reusable molecule/group definitions for molecular liquids, polymers, and flexible adsorbates.

Interface profiles

Add analysis-owned one-dimensional number-density, mass-density, composition, orientation, and stress profiles along a user-declared interface coordinate. The interface normal and region assignment must be explicit inputs.

Porous and framework-specific extension

Primitive rings, ring centers, windows, cages, natural tilings, and site semantics are not owned here. They are optional structural extensions governed by:

- `framework_ring_architecture.{md,pdf}`;
- `topology_statistics_architecture.{md,pdf}`;
- `stagell_site_kinetics_architecture.{md,pdf}`.

An MLFF material profile may append those calls when the system has a meaningful porous or framework topology. Generic material profiles must not activate them.

Testing and acceptance

Every new structural observable requires:

1. analytic or exactly enumerable synthetic tests;
2. triclinic and periodic-boundary tests;
3. selection and empty-data failure tests;
4. backend-equivalence tests where neighbor search is used;
5. immutable result and provenance tests;
6. at least one physically interpretable domain fixture;
7. a separate module specification before MLFF exposes it through the call registry.

Thermomechanical and energetic boundary

This manual does not own equation-of-state fitting, elastic constants, thermodynamic fluctuation response, stress-correlation viscosity, harmonic phonons, surface/interface energies, defect energetics, or migration barriers. Those analyses are defined in `thermomechanical_energetic_validation_architecture.{md,pdf}`. Structural observables may be used as diagnostics for those workflows, but their numerical definitions and result schemas remain here.